THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): The temperature in the stratosphere rises with height. Reason (R): The ozone layer absorbs ultraviolet rays from the Sun. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Two plane mirrors are at right angles. A ray hits the first at an angle of incidence of 25 degrees. The angle of incidence on the second is: 1

(a) 65 degrees **(b)** 50 degrees **(c)** 25 degrees **(d)** 90 degrees

3. A student measures the shadow of a 1 m stick as 0.5 m at noon on one day and 1 m a few months later. This shows the Sun's noon elevation has: 1

(a) become zero **(b)** stayed same **(c)** increased **(d)** decreased

4. Producers form the broad base of an ecosystem pyramid because they: 1

(a) do not need sunlight
(b) are the largest animals
(c) support all the other levels and hold the most energy
(d) eat the consumers

5. If the volume of water in oceans is about 1.35 billion cubic km and total fresh water is 3 per cent of total water, with total water about 1.4 billion cubic km, the fresh water volume is about: 1

(a) 420 thousand cubic km **(b)** 4.2 billion cubic km
(c) 42 million cubic km **(d)** 1.4 million cubic km

6. A torch bulb is placed at the focus of a concave reflector. The beam leaving the torch is: 1

(a) convergent **(b)** parallel
(c) divergent **(d)** scattered in all directions

7. A calendar with 12 months of 29.5 days is adjusted by adding a month every: 1

(a) about 3 years **(b)** every 10 years **(c)** every year **(d)** every month

8. The relationship between a tapeworm and its host is: 1

(a) competition **(b)** predation **(c)** mutualism **(d)** parasitism

9. Which is the most effective individual action to reduce carbon dioxide in the atmosphere? 1

(a) using more fuel
(b) cutting trees
(c) burning waste
(d) using public transport and saving electricity

10. Assertion (A): Convex mirrors are used as rear-view mirrors in vehicles. Reason (R): They always form real and enlarged images. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Assertion (A): We always see the same side of the Moon. Reason (R): The Moon does not rotate on its axis. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

12. Assertion (A): Food chains rarely have more than four or five trophic levels. Reason (R): Most of the energy is lost as heat at each trophic level. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

13. **Assertion (A):** The outer core of the Earth is liquid. 1
Reason (R): The outer core consists of molten iron and nickel at very high temperature.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** Convex mirrors are used as rear-view mirrors in vehicles. 1
Reason (R): A convex mirror gives a wider field of view.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** We always see the same side of the Moon from the Earth. 1
Reason (R): The Moon takes the same time to rotate once on its axis as to revolve once around the Earth.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. State the reason why the Moon cannot be seen at the time of a new moon, although it is above the horizon in the daytime. 2

17. How do forests act as carbon sinks? 2

18. Why is the exosphere considered the boundary between the atmosphere and space? 2

19. Why is a convex lens also called a converging lens, and where does a beam parallel to its axis meet? 2

20. How did the discovery of the Earth's revolution explain the changing night sky through the year? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain how the Earth's atmosphere developed over time from the early atmosphere to the oxygen-rich present. 3

22. Compare real image formation in a camera and an eye in terms of lens and screen. 3

23. Explain the length of the Earth's orbit and the speed of the Earth in orbit (use 150 million km radius, circumference 2 pi r) in km/s approx. 3

24. Why do some scientists recommend reducing meat consumption? Use energy ideas. 3

OR

Explain how the Earth's rotation would change the length of day and night if its speed doubled, and its effect on life. 3

25. An object is 15 cm in front of a plane mirror. Find the distance of the image from the mirror and from the object. If the object moves 4 cm towards the mirror, find the new distance between object and image. 3

OR

A year consists of 365.2422 days. Calculate how many days the calendar will be ahead after 100 years if 365.25 is used and say how the Gregorian rule corrects it. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Kavya fixes a vertical stick in the playground of her school in Nagpur and marks the tip of its shadow every hour from 8 a.m. to 4 p.m. She finds that the shadow is longest in the morning and evening and shortest around midday. 5

(i) In which direction does the shadow point in the morning?
(ii) At what time of day is the shadow shortest?
(iii) Name the device that works on this principle.
(iv) Explain why the shadow length changes during the day.

27. Discuss how rising CO2 in the atmosphere affects plants, oceans, ice and weather, and what individuals can do. 5

OR

Compare the effects of increased CO2 on the oceans (acidification), the atmosphere (warming) and plants (growth). Discuss the benefits and the harms. 5

END OF QUESTION PAPER — Total: 50 Marks